origin in some extra-terrestrial source. Test equipment carried outside of the earth's atmosphere by rockets has given us data which provide a fairly accurate picture of the composition of the radiation that comes to the earth from space.

The investigation of cosmic radiation has had a profound influence on nuclear science. When Chadwick in 1932 discovered the neutron, the picture of matter seemed complete: all matter appeared to be composed of four fundamental particles: protons, neutrons, electrons, and photons. The existence of the electron neutrino was predicted by Pauli in 1927 but it was not proven until 1956. However, through studies of the cosmic radiation Anderson discovered the *positron* (the first *antiparticle*) in the same year. Five years later Anderson and Neddermeyer discovered another new particle with a mass about one-tenth of a proton or about 200 times heavier than the electron. This particle is the *muon*, designated by μ. Since that time a large number of subatomic particles have been discovered.

13.1. PRIMARY COSMIC RADIATION

A rather small fraction of the cosmic radiation consists of electromagnetic radiation and electrons. The former vary in energy from a small percentage of γ-rays to a considerable intensity of X-rays, to visible light and to radiation in the radiofrequency region. The types and intensities of this radiation have been of great importance to development of models of the formation and composition of the universe.

The major part of the cosmic radiation is nuclear particles with very high energy: approximately 70% protons, 20% α-particles, 0.7% lithium, beryllium, and boron ions, 1.7% carbon, nitrogen, and oxygen ions, the residual 0.6% ions of $Z > 10$. These ions are bare nuclei prior to interaction since their kinetic energies exceed the binding energies of all of the orbital electrons.

The cosmic particle radiation can be divided by energy into two major groups (Fig. 13.1). One group has energies mainly below 1 GeV and consists primarily of protons. This group originates mainly from the sun. Its intensity varies in relation to solar eruptions since at the time of such an eruption a large amount of solar material, primarily hydrogen, is ejected into space.

The second group has energies up to about 3×10^{11} GeV, although the intensity of the particles decreases with increasing energy, following the relation $N(E) \propto E^{-1.6}$, where $N(E)$ is the number of particles with energies in excess of E. Thus particles of 10^3 GeV have an intensity of about 10^{11} higher than particles of 10^{10} GeV. Within this high energy group the particles at the lower end of the energy spectrum are assumed to originate from sources within our galaxy (the Milky Way), while the particles of the higher energy end are assumed to come from sources outside of our galaxy. Different hypotheses, which for the most part are untested, suggest that the particles come from astronomical radio sources, exploding super novae, or colliding galaxies, etc. At least a

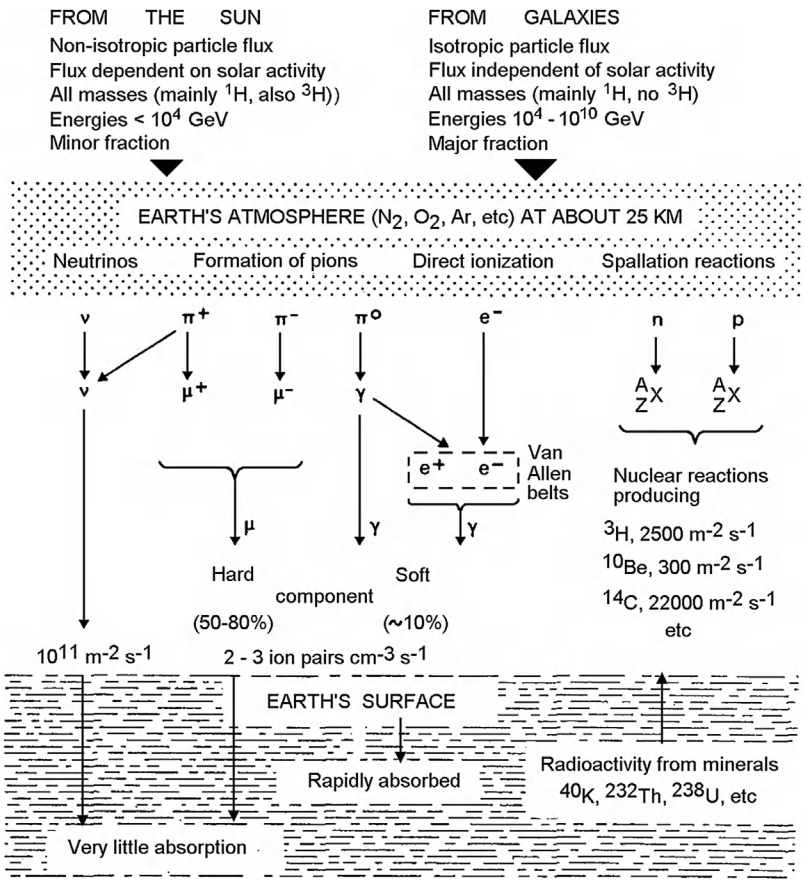


Figure 13.1 Cosmic radiation consists of atoms and photons which react with the atmosphere leading to the formation of numerous secondary particles, some (but not all) detectable at the Earth's surface.

portion of this radiation is a residue of the processes involved in the original formation of the universe. It is assumed that the high energy particles obtain their tremendous kinetic energies through acceleration in the magnetic field of galactic objects (synchrotron acceleration, §16.6).

When the primary cosmic particles enter the earth's atmosphere, they collide with the matter of the atmosphere and are annihilated. In this annihilation process a large number of new particles are formed whose total kinetic energy is less than that of the original primary radiation but whose total rest mass is larger than that of the primary particle. A 10^{4} GeV cosmic helium ion may produce a shower of 50—100 new highly ionizing particles, cf. Fig. 2.1. The main reaction products are particles, which are known as pions, designated π.

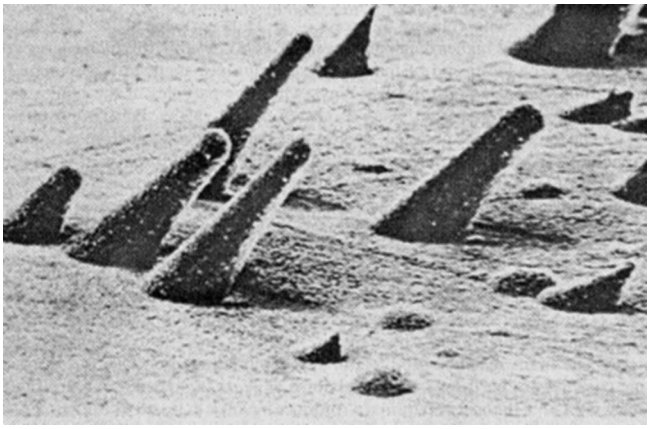


Figure 13.2 Holes in helmet of Apollo 12 astronaut caused by high energy cosmic rays. The holes have been made visible in a microscope by etching *(From New Scientist, April 22, 1973).*

Figure 13.2 shows the effect of high energy cosmic rays hitting the helmets of Apollo 12 astronauts. It is probable that the cosmic ray intensity will put a limit to how long man can endure in outer space: it has been calculated that in a journey to the planet Mars about 0.1% of the cerebral cortex will be destroyed. The annihilation process occurs to such an extent that below an altitude of approximately 25 km above the earth the number of primary cosmic particles has been reduced to quite a small fraction of the original intensity.

The discovery of π-mesons (or *pions*) was reported by Powell and Occhialini in 1948 after they had analyzed tracks in photographic emulsions placed for some months on a mountain top to get a high yield of cosmic ray interactions. Pions are produced in large amounts in all high energy ($\geq$400 MeV) nuclear reactions. In 1935 Yukawa suggested that the nucleons in a nuclide were held together through the exchange of a hypothetical particle, which we now recognize as the pion, just as hydrogen atoms in H_2 are held together through the exchange of an electron:

$$p \leftrightarrow n + \pi^+; \quad n \leftrightarrow p + \pi^-; \quad p \overset{\pi^\circ}{\leftrightarrow} p; \quad n \overset{\pi^\circ}{\leftrightarrow} n; \tag{13.1}$$

Pions are the particles of *strong interaction,* cf. §2.2. Since the pion rest mass is 0.147 u, the energy to produce a pion is at least 137 MeV (in practice, in order to conserve momentum it exceeds 400 MeV, cf. §4.6 and 5.7). The pions are unstable in free form.

13.2. SECONDARY REACTIONS IN THE EARTH'S ATMOSPHERE

Few of the pions formed in the annihilation process reach the earth's surface. They undergo radioactive decay (life-time about 10^{-6} s) to muons and muon neutrinos